





গণপ্রজাতন্ত্রী বাংলাদেশ





x

1

7

x

10

0

1

0

1







PARADOXES

ESPRESSO

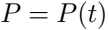




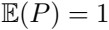




BEAR







100%







2023-10-21

$$= \underbrace{\frac{p_t^2}{2 \log(2)} \pi \lambda \left(\frac{h \varphi_{\text{RX}}}{\sin^2(\epsilon)} \right)^2}_{k \triangleq} \exp \left\{ - \tau^2 \underbrace{v_{\text{sat}}^2 \sin^4(\epsilon) / (h^2 \varphi_{\text{RX}}^2)}_{D \triangleq} \log(2) / 2 \right\},$$





























Fig 1(a): True $P(t)$, GPR estimate & forecast, and 100-sample MA & forecast

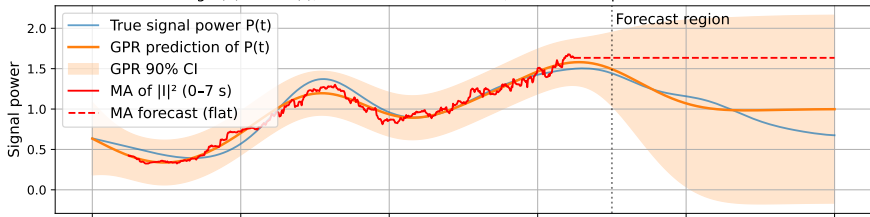
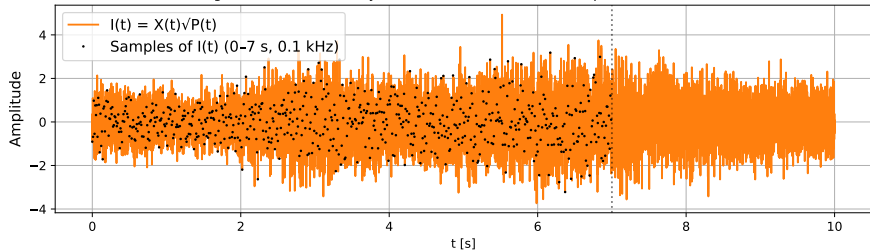


Fig 1(b): Non-stationary Gaussian noise $I(t)$ with sampled observations









expanding world + new world







Qatar Airways









$$S_{RX}(t) = \frac{A}{l(d_0)} \sqrt{\frac{K}{K+1}} e^{-2i\pi\tau_0(t)f_c} e^{-2i\pi\theta} + \sum_{j=1}^{100} \frac{A}{l(d_j) \sqrt{100} \sqrt{K+1}} e^{-2i\pi\tau_j(t)f_c} e^{-2i\pi\theta},$$

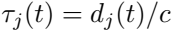


















QWERTY

$$v = \sqrt{\frac{GM}{h + R_{\oplus}}},$$

WORLDWIDE







20

—

12

.

100



$$T_c = \frac{1}{8D_g} \approx 10^{-3}$$





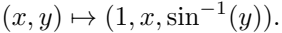


















$$1 = e^{-\frac{\pi^2}{2\sqrt{P}}}$$



$$] = 1 - \left(1 - e^{-\frac{\pi^2}{2\sqrt{P/N}}} \right)^N,$$





192







$$t \rightarrow \text{order} \rightarrow \frac{\sin(2\pi B t)}{\pi t}$$





$$\mathbb{Z} \ni i \mapsto \frac{\sin\left(2\pi \frac{B_L}{f_c} i\right)}{\pi i},$$

$$i \mapsto \delta[i] - \frac{\sin(2\pi \frac{B_H}{f_c} i)}{\pi i},$$



WAVELENGTH



WINNERS

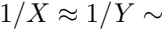
$$\mu = \exp\left(\mu_{\text{LN}} + \frac{\sigma_{\text{LN}}^2}{2}\right)$$

$$O(2\pi) \cong SO(2) \cong \mathbb{R}/2\pi\mathbb{Z}$$



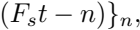






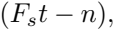
FOR THE WIN





1991-10-1

$$S(t) = \sum_{n=0}^{N-1} x[n]$$













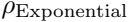


$$H = \begin{pmatrix} -a & -0 & -1 & 0 & 0 & 0 & ar & 0 & r \\ 0 & 0 & 0 & -a & -0 & -1 & 0 & 0 & 0 \\ -c & -b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & -b^2/a & -1 & cr & rb^2/a & r \\ a & 0 & -1 & 0 & 0 & 0 & ar & 0 & -r \\ 0 & 0 & 0 & a & 0 & -1 & 0 & 0 & 0 \\ -c & b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & b^2/a & -1 & -rc & rb^2/ar & -r \end{pmatrix}$$

$$\lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X \geq x)}{x},$$







$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{-x/\theta}{x} = 1/\theta.$$



$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(1 - \gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$\begin{aligned}
 &\stackrel{(a)}{=} \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta))}{x} = \lim_{x \rightarrow \infty} - \frac{\log(x^{k-1} e^{-x/\theta})}{x}
 \end{aligned}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x} = 1/\theta.$$



$$\lim_{x \rightarrow \infty} \frac{\Gamma(s, x)}{x^{s-1} e^{-x}} = 1.$$







$$\begin{aligned}
&= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = - \lim_{x \rightarrow \infty} \frac{\log \operatorname{erfc} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)}{x} \stackrel{(b)}{\geq} - \lim_{x \rightarrow \infty} \frac{\log \left(\sqrt{\frac{e}{2\pi}} e^{-2 \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)^2} \right)}{x}
\end{aligned}$$

$$\geq -\lim_{x \rightarrow \infty} \frac{\log \left(e^{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2} \right)}{x} = -\lim_{x \rightarrow \infty} \frac{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2}{x} = \frac{1}{\sigma^2} \lim_{x \rightarrow \infty} \frac{x^2 - 2x\mu + \mu^2}{x} = \infty,$$



$$\phi(x) \geq \sqrt{\frac{2e}{\pi}} \frac{\sqrt{\beta-1}}{\beta} e^{-\beta x^2},$$















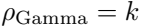
























Q1 2024

2023年12月



$$\mathbb{E}[(S_1 + S_2)^2] = \int_0^1 (\cos(2\pi t) + \cos(2\pi t + \pi))^2 dt = \int_0^1 0 dt = 0,$$



1921 + 1921

$$= \int_0^1 \cos^2(2\pi t) dt + \int_0^1 \cos^2(2\pi t + \pi) dt$$

$$= \int_0^1 2 \cos^2(2\pi t) dt = \int_0^1 \cos(4\pi t) dt + 1 = 1.$$





$$E[S_1^2 + S_2^2] = E[S_1^2] + E[S_2^2] = E[S_1^2] + 2E[S_1 S_2] + E[S_2^2]$$

1991 + 1991 + 1991

1992-93

$$\mathbb{E}[S_1 S_2] = \int_0^1 \cos(2\pi t) \cos(2\pi t + \pi) dt = \int_0^1 -\cos^2(2\pi t) dt = -\frac{1}{2} \int_0^1 \cos(4\pi t) dt - \frac{1}{2} = -\frac{1}{2},$$

01

,

02

E

1

0

,

2

1

1

2024 + 2025

2023-04-20

$$\mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1^2 + S_2^2]] = \mathbb{E} \phi_1 [\mathbb{E} [S_1^2]] + \mathbb{E} \phi_2 [\mathbb{E} [S_2^2]] + \mathbb{E} \phi_1 \phi_2 [\mathbb{E} [S_1 S_2]]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2$$

$$+ \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \int_0^1 \cos(2\pi t + \phi_1) \cos(2\pi t + \phi_2) dt d\phi_1 d\phi_2$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2 = 1/2 + 1/2 = 1.$$

1992

2020-2021





$${}_3F_2(1, 1, 1 + b; 2, 2; x) = \sum_{n=0}^{\infty} \frac{(1)_n (1)_n (1 + b)_n}{(2)_n (2)_n} \frac{x^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(1+b)_n}{(n+1)^2 n!} x^n = \frac{1}{b!} \sum_{n=0}^{\infty} \frac{(n+1)_b}{(n+1)^2} x^n$$

$$\underline{\underline{(a)}} \quad \frac{1}{b!} \sum_{n=0}^{\infty} \frac{\sum_{k=1}^b \begin{bmatrix} b \\ k \end{bmatrix} (n+1)^k}{(n+1)^2} x^n$$

$$\begin{aligned}
 &= \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \sum_{n=0}^{\infty} \frac{x^n}{(n+1)^{2-k}} \stackrel{(b)}{=} \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \frac{\operatorname{Li}_{2-k}(x)}{x},
 \end{aligned}$$

b

16

